

Foundations of CycleGAN

[CycleGAN]

1.0 Content Block

Move order and strategic equilibria In the original paper, as well as most subsequent papers, it is usually assumed that the generator moves first, and the discriminator moves second, thus giving the following minimax game: $\min_{\mu_G} \max_{\mu_D} L(\mu_G, \mu_D) := \mathbb{E}_{x \sim \mu_{\text{ref}}, y \sim \mu_D(x)} [\ln \mu_y] + \mathbb{E}_{x \sim \mu_G, y \sim \mu_D(x)} [\ln (1 - y)]$.
$$\min_{\mu_G} \max_{\mu_D} L(\mu_G, \mu_D) := \mathbb{E}_{x \sim \mu_{\text{ref}}, y \sim \mu_D(x)} [\ln y] + \mathbb{E}_{x \sim \mu_G, y \sim \mu_D(x)} [\ln(1-y)].$$

2.0 Content Block

A generative adversarial network (GAN) is a class of machine learning frameworks and a prominent framework for approaching generative artificial intelligence. The concept was initially developed by Ian Goodfellow and his colleagues in June 2014. In a GAN, two neural networks compete with each other in the form of a zero-sum game, where one agent's gain is another agent's loss. Given a training set, this technique learns to generate new data with the same statistics as the training set. For example, a GAN trained on photographs can generate new photographs that look at least superficially authentic to human observers, having many realistic characteristics. Though originally proposed as a form of generative model for unsupervised learning, GANs have also proved useful for semi-supervised learning, fully supervised learning, and reinforcement learning. The core idea of a GAN is based on the "indirect" training through the discriminator, another neural network that can tell how "realistic" the input seems, which itself is also being updated dynamically. This means that the generator is not trained to minimize the distance to a specific image, but rather to fool the discriminator. This enables the model to learn in an unsupervised manner. GANs are similar to mimicry in evolutionary biology, with an evolutionary arms race between both networks.

3.0 Content Block

Nash equilibrium $(\hat{\mu}_D, \hat{\mu}_G)$
$$\{\hat{\mu}_D, \hat{\mu}_G\}$$
, which is stable under simultaneous move order: $\hat{\mu}_D \in \arg \max_{\mu_D} L(\mu_G, \mu_D)$, $\hat{\mu}_G \in \arg \min_{\mu_G} L(\mu_G, \hat{\mu}_D)$
$$\hat{\mu}_D \in \arg \max_{\mu_D} L(\hat{\mu}_G, \mu_D), \hat{\mu}_G \in \arg \min_{\mu_G} L(\mu_G, \hat{\mu}_D)$$

4.0 Content Block

Bidirectional GAN (BiGAN) The standard GAN generator is a function of type $G: \Omega_Z \rightarrow \Omega_X$
$$G: \Omega_Z \rightarrow \Omega_X$$
, that is, it is a mapping from a latent space Ω_Z
$$\Omega_Z$$
 to the image space Ω_X
$$\Omega_X$$
. This can be understood as a "decoding" process, whereby every latent vector $z \in \Omega_Z$
$$z \in \Omega_Z$$
 is a code for an image $x \in \Omega_X$
$$x \in \Omega_X$$
, and the generator performs the decoding. This naturally leads to the idea of training another network that performs "encoding", creating an autoencoder out of the encoder-generator pair. Already in the original paper, the authors noted that "Learned approximate inference can be performed by training an auxiliary network to predict z
$$z$$
 given x
$$x$$
". The bidirectional GAN architecture performs exactly this.

5.0 Content Block

Knight, Will. "5 Big Predictions for Artificial Intelligence in 2017". MIT Technology Review. Retrieved January 5, 2017. Karras, Tero; Laine, Samuli; Aila, Timo (2018). "A Style-Based Generator Architecture for

Generative Adversarial Networks". arXiv:1812.04948 [cs.NE]. This Person Does Not Exist – photorealistic images of people who do not exist, generated by StyleGAN This Cat Does Not Exist Archived March 5, 2019, at the Wayback Machine – photorealistic images of cats who do not exist, generated by StyleGAN Wang, Zhengwei; She, Qi; Ward, Tomas E. (2019). "Generative Adversarial Networks in Computer Vision: A Survey and Taxonomy". arXiv:1906.01529 [cs.LG].

6.0 Content Block

(Ω_Z, μ_Z) , the latent space. There are 3 players in 2 teams: generator, encoder, and discriminator. The generator and encoder are on one team, and the discriminator on the other team. The generator's strategies are functions $G : \Omega_Z \rightarrow \Omega_X$, and the encoder's strategies are functions $E : \Omega_X \rightarrow \Omega_Z$. The discriminator's strategies are functions $D : \Omega_X \rightarrow [0, 1]$. The objective function is $L(G, E, D) = \mathbb{E}_{x \sim \mu_X} [\ln D(x, E(x))] + \mathbb{E}_{z \sim \mu_Z} [\ln (1 - D(G(z), z))]$

7.0 Content Block

The discriminator's strategy set is the set of measurable functions of type $D : \Omega \rightarrow \mathbb{R}$ with bounded Lipschitz norm: $\|D\|_L \leq K$, where K is a fixed positive constant. The objective is $L_{WGAN}(\mu_G, D) := \mathbb{E}_{x \sim \mu_G} [D(x)] - \mathbb{E}_{x \sim \mu_{ref}} [D(x)]$

8.0 Content Block

The discriminator's strategy set is the set of Markov kernels $\mu_D : (\Omega, \mathcal{B}) \rightarrow \mathcal{P}([0, 1], \mathcal{B}([0, 1]))$, where $\mathcal{B}([0, 1])$ is the Borel σ -algebra on $[0, 1]$. Since issues of measurability never arise in practice, these will not concern us further.

9.0 Content Block

Concerns have been raised about the potential use of GAN-based human image synthesis for sinister purposes, e.g., to produce fake, possibly incriminating, photographs and videos. GANs can be used to generate unique, realistic profile photos of people who do not exist, in order to automate creation of fake social media profiles. In 2019 the state of California considered and passed on October 3, 2019, the bill AB-602, which bans the use of human image synthesis technologies to make fake pornography without the consent of the people depicted, and bill AB-730, which prohibits distribution of manipulated videos of a political candidate within 60 days of an election. Both bills were authored by Assembly member Marc Berman and signed by Governor Gavin Newsom. The laws went into effect in 2020. DARPA's Media Forensics program studies ways to counteract fake media, including fake media produced using GANs.